

Subgroup Analysis of Phase 3 Study of FDC of Dapagliflozin, Glimepiride, and Metformin Extended Release in Type 2 Diabetes Mellitus Patients with Mild Renal Impairment

Abstract Number:
A-1-144362-RSSDI2023

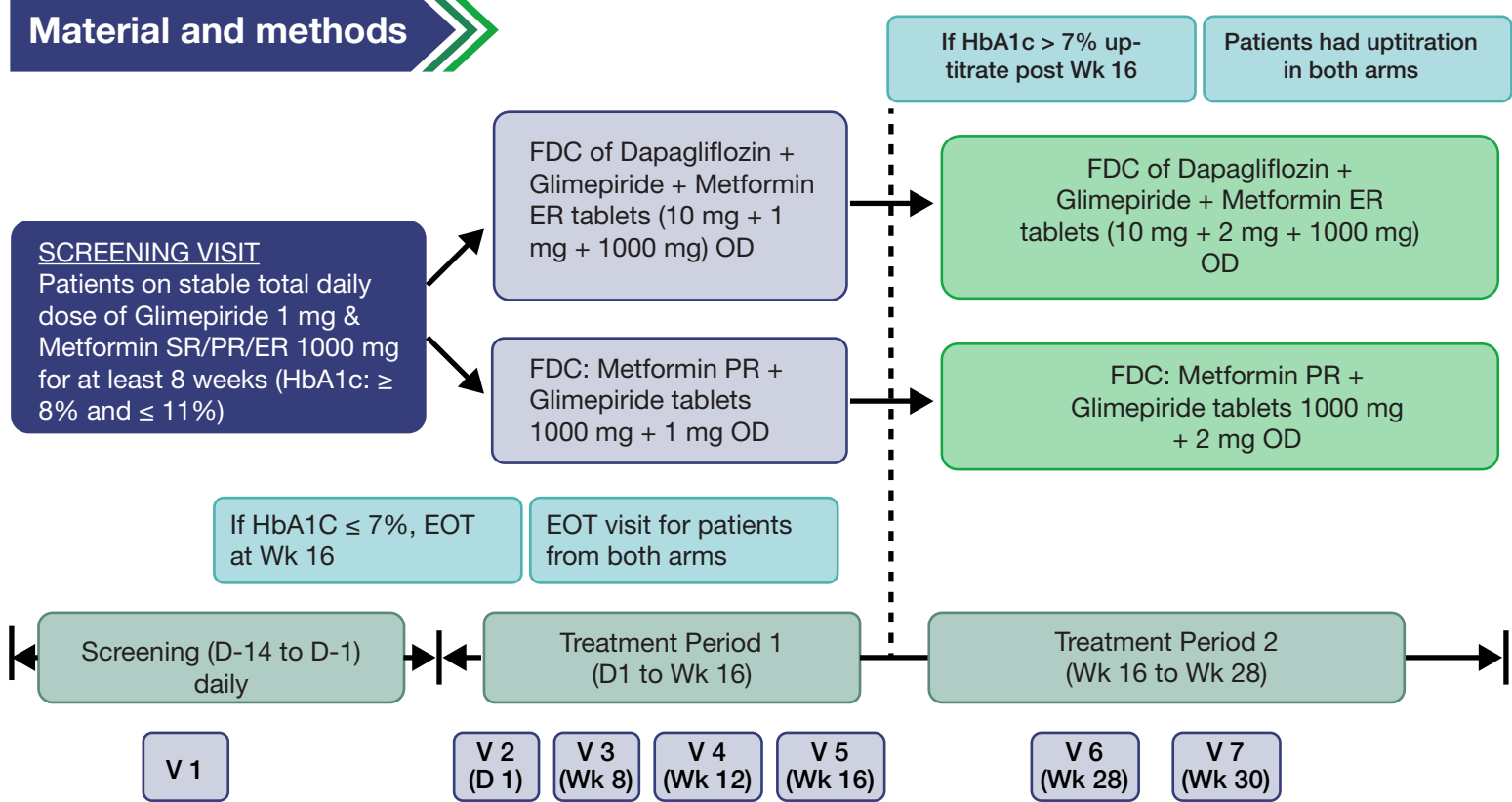
Patil P^{1*}, Sahay R², Agarwal S³, Agarwal V⁴, Rai P⁵, Sharma S⁶, Pujara G⁷, Murthy S⁸, Saha S¹, Patel P¹, Patil D¹, Ghadge P¹, Lakhwani L¹, Mehta S¹, Joglekar S⁹

*Presenting author: 1. Sun Pharma Laboratories Ltd, Mumbai, India; 2. Osmania Medical College, Hyderabad, India; 3. GSVM Medical College, Kanpur, India; 4. Surya Super Speciality Hosp, Varanasi, India; 5. Opal Hospital, Varanasi, India; 6. Diabetes Thyroid & Endocrine Centre, Jaipur, India; 7. Maruti Multispecialty Hospital, Ahmedabad, India; 8. Life Care Hospital, Bangalore, India; 9. Sun Pharmaceutical Industries Ltd, Mumbai, India. Dapagliflozin, Glimepiride and Metformin extended release Study group: M Srivastava, P Kurmi, J Ambaliya, N Lomte, S Gofne, S Agarwal, P Kashid, V Agarwal, H

Background and Aim

- Up to 40% of patients with T2DM eventually suffer from kidney failure.^{1,2} South Asians have a higher disposition for CKD and a 4-fold more likelihood of developing albuminuria.³
- In T2DM, HbA1c < 7% (< 53 mmol/mol) reduces the risk of overall microvascular and macrovascular complications.^{4,5}
- SGLT2 inhibitors such as Dapagliflozin has established cardio-renal protective effect in patients with T2DM.^{6,7}
- Subgroup analysis aimed to assess efficacy and safety of triple-drug FDC vs dual-drug FDC in T2DM patients with mild renal impairment (baseline eGFR [60 to <90 mL/min/1.73 m²]).

Material and methods



Study design

A phase III, randomized, open-label, two-arm, multicenter, parallel-group, active-controlled comparative study

Primary endpoint

Mean change in HbA1c from baseline to Week 16

Secondary endpoints

- Proportion of patients achieving HbA1c < 7.0%
- Mean change in PPBG and FBG from baseline to Week 16
- Incidence of TEAEs
- Patients requiring hypoglycemia management

Key inclusion criteria

- T2DM patients of either gender, aged 18–65 years
- HbA1c ≥ 8.0% – ≤ 11%
- Lifestyle modifications and stable daily dose of Glimepiride 1 mg and Metformin SR/PR/ER 1000 mg for at least 8 weeks prior to screening

Key exclusion criteria

- FBG ≥ 270 mg/dL at screening
- Any weight loss medications within 3 months prior to randomization
- Significant renal impairment (eGFR < 45 mL/min/1.73 m²) or hepatic impairment (AST and ALT > 3 times upper limit of normal)
- Uncontrolled hypertension ≥ 160/100 mmHg

Study arms

- Test arm: FDC of Dapagliflozin + Glimepiride + Metformin ER tablets (10 mg + 1 mg + 1000 mg) OD
- Comparator arm: FDC of Metformin PR + Glimepiride tablets (1000 mg + 1 mg) OD

Results

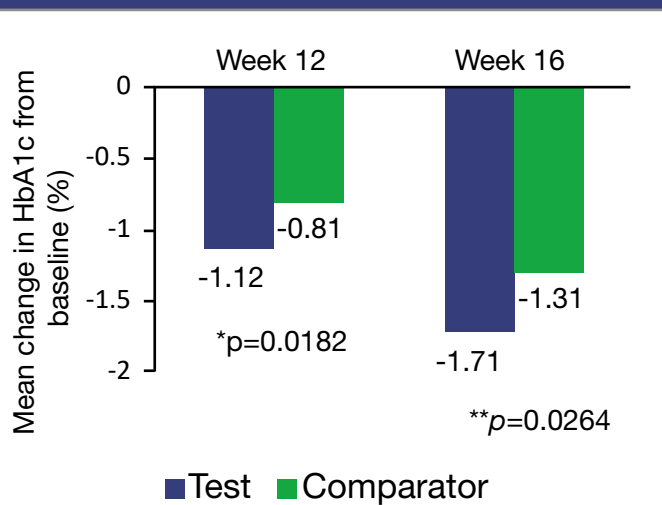
Table 1: Demographic and baseline characteristics

	Test arm (N = 41)	Comparator arm (N = 45)
Gender		
Male, n (%)	20 (48.8%)	21 (46.7%)
Female, n (%)	21 (51.2%)	24 (53.3%)
Age, Mean ± SD	52.32 ± 9.22	50.60 ± 9.44
Weight (kg), Mean ± SD	67.57 ± 16.77	66.56 ± 11.63
BMI, (kg/m ²), Mean ± SD	25.96 ± 4.91	26.58 ± 4.68
HbA1c (%), Mean ± SD	8.89 ± 0.91	8.80 ± 0.75
FBG (mg/dL), Mean ± SD	237.7 ± 55.11	254.2 ± 56.53
PPBG (mg/dL), Mean ± SD	173.2 ± 30.29	171.6 ± 38.93

Change in HbA1c levels

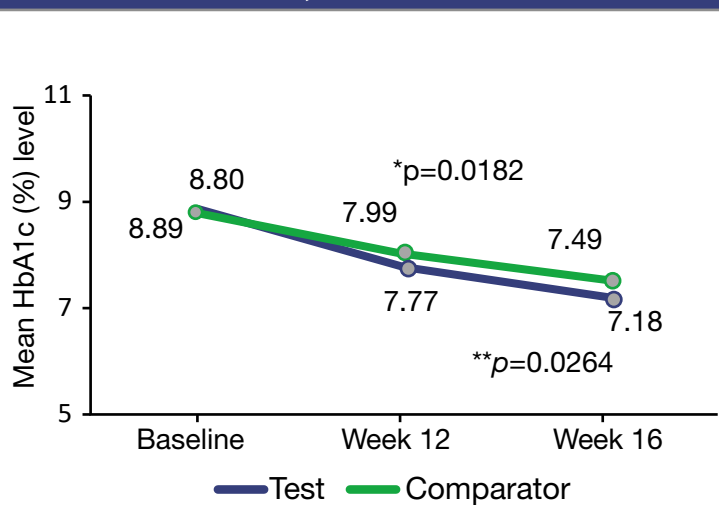
Reduction in HbA1c from baseline to week 16 demonstrated superior efficacy in test vs. comparator arm at Week 12 ($p=0.0182$) and at Week 16 ($p=0.0264$), Figure 1.

Figure 1: Mean change in HbA1c from baseline to Weeks 12 and 16



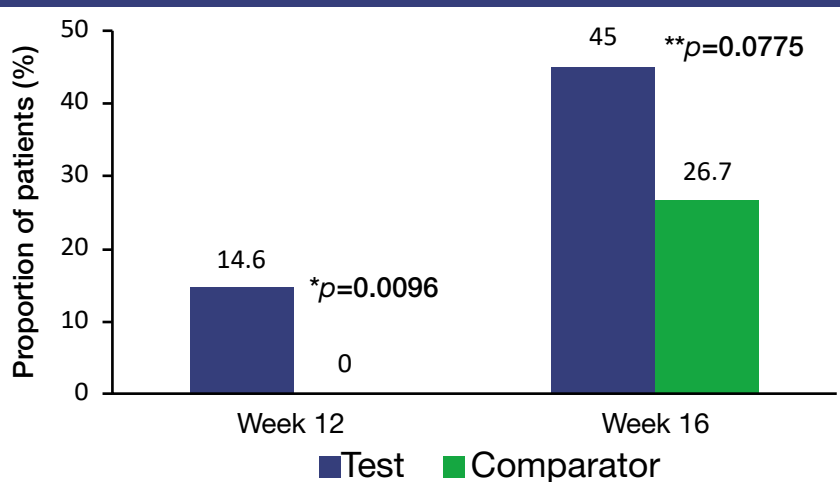
*Test vs. comparator at Week 12. **Test vs. comparator at Week 16.

Figure 2: Mean HbA1c levels at baseline, Week 12 and 16



*Test vs. comparator at Week 12. **Test vs. comparator at Week 16.

Figure 3: Patients achieving HbA1c < 7% in test vs. comparator arm

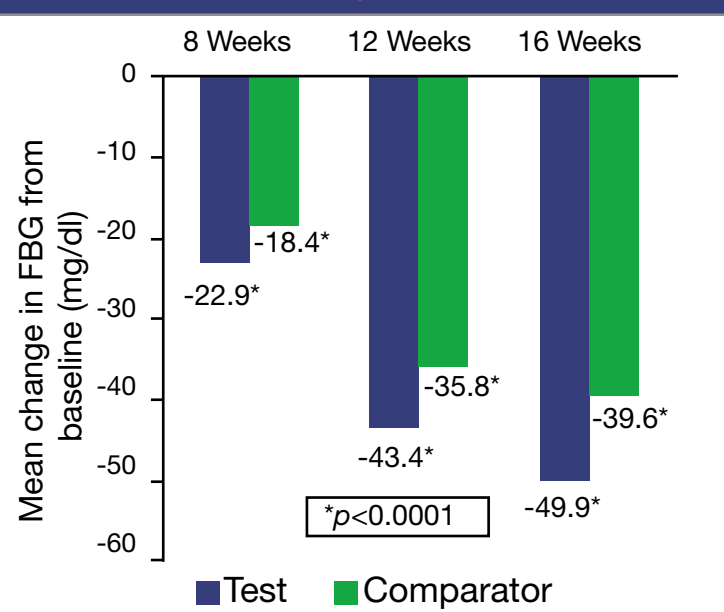


*Test vs. comparator at Week 12. **Test vs. comparator at Week 16.

Mean change in FBG and PPBG

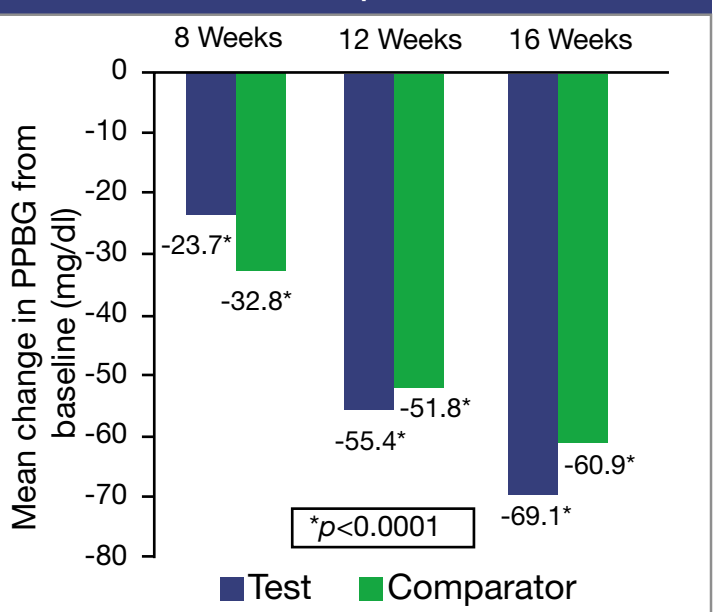
Significant reduction in FBG and PPBG was observed within test and comparator arms at Week 8, 12 and 16 ($p<0.0001$) from baseline. Reduction in FBG and PPBG was comparable between both arms at Weeks 8, 12 and 16

Figure 4: Change in FBG levels in test vs. comparator arm



*Change from baseline within group

Figure 5: Change in PPBG levels in test vs. comparator arm



*Change from baseline within group

Safety

- Only one TEAE of hypoglycemia which was mild in nature was reported in test arm and did not require intervention. The event was resolved and dose of study drug was not changed
- No TEAE leading to study drug interruption, discontinuation or dose reduction was reported during the study period. No SAEs were reported
- None of the patients required rescue medication
- All treatments were well-tolerated

Conclusion

As per the results of subgroup analysis, triple FDC of Dapagliflozin + Glimepiride + Metformin hydrochloride ER had a superior efficacy in terms of HbA1c reduction and comparable safety profile to dual FDC of Metformin PR + Glimepiride in T2DM patients with mild renal impairment. Both the study treatment was found to be well-tolerated.

Abbreviations:

ALT, Alanine transaminase; AST, Aspartate aminotransferase; BMI, Body mass index; CKD, Chronic kidney disease; D, Day; EOT, End of treatment period 1 (week 16); eGFR, Estimated glomerular filtration rate; ER, Extended release; FBG, Fasting blood glucose; FDC, Fixed dose combination; HbA1c, Glycated haemoglobin; OD, Once daily; PPBG, Postprandial blood glucose; PR, Prolonged release; SAE, Serious adverse event; SGLT2, Sodium-glucose cotransporter-2; SR, Sustained release; T2DM, Type 2 diabetes mellitus; TEAE, Treatment emergent adverse event; V, Visit; Wk, Week

References:

1. Thomas MC, et al. Changing epidemiology of type 2 diabetes mellitus and associated chronic kidney disease. *Nat Rev Nephrol*. 2016 Feb;12(2):73–81. 2. USRDS: the United States Renal Data System. *Am J Kidney Dis*. 2003 Dec;42(6 Suppl 5):1–230. 3. Shah A, et al. Diabetes and associated complications in the South Asian population. *Curr Cardiol Rep*. 2014 May;16(5):476. 4. Kidney Disease: Improving Global Outcomes (KDIGO) Diabetes Work Group. KDIGO 2022 Clinical practice guideline for diabetes management in chronic kidney disease. *Kidney Int*. 2022 Nov;102(S5):S1–S127. 5. Ioannidis I. Diabetes treatment in patients with renal disease: Is the landscape clear enough? *World J Diabetes*. 2014 Oct 15;5(5):651–658. 6. Cahn A, et al. Cardiovascular, renal, and metabolic outcomes of dapagliflozin versus placebo in a primary cardiovascular prevention cohort: analyses from DECLARE-TIMI 58. *Diabetes Care*. 2021 May;44(5):1159–1167. 7. de Boer IH, et al. Diabetes management in chronic kidney disease: a consensus report by the American diabetes association (ADA) and kidney disease: improving global outcomes (KDIGO). *Diabetes Care*. 2022 Dec 1;45(12):3075–3090.